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Inventor(s)	Contreras Sanchez; Alberto

Multipurpose protective cap for transmission drive shaft assembly

Abstract

A multipurpose protective cap for a transmission prior to and during assembly with a drive shaft has a base portion having a cylindrical cross-section of a first inner diameter. The base portion comprises a side wall defining the cylindrical cross-section and having an outer edge and an inner edge, and a membrane spanning the cylindrical cross-section of the base portion at the outer edge of the side wall. An insertion portion extends from the base portion opposite the outer edge, the insertion portion having a cylindrical cross-section of a second inner diameter, the second inner diameter smaller than the first inner diameter. A contamination removal portion is formed in an interior surface of the insertion portion and configured to remove debris as the drive shaft passes through the insertion portion. A handle extends from the side wall of the base portion.

Inventors:	Contreras Sanchez; Alberto (Toluca, MX)
Applicant:	Nissan North America, Inc. (Franklin, TN)
Family ID:	1000008779766
Assignee:	Nissan North America, Inc. (Franklin, TN)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1271077	12/1917	Probasco	277/499	F16J 9/14
1398439	12/1920	McMaster	277/497	F16J 9/14
4218813	12/1979	Cather, Jr.	29/464	B25B 27/0028
4902043	12/1989	Zillig	285/379	F16L 37/0987
6632999	12/2002	Sempliner	52/27	F16L 5/00
2011/0179620	12/2010	Harmon	49/463	F16J 15/3268

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1240980	12/2001	EP	B25B 27/0028

OTHER PUBLICATIONS

Translation of EP 1240980 A1, obtained from FIT database (Year: 2002). cited by examiner

Primary Examiner: Diaz; Thomas C

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to a multipurpose protective cap used to protect the transmission prior to assembly with the drive shaft, protect the oil seal ring during insertion of the drive shaft and remove debris from the drive shaft during insertion into the transmission.

BACKGROUND

(2) Currently, transmissions are stored and/or transported prior to assembly with a cap that seals the opening in the transmission configured to receive the drive shaft. The drive shaft is stored and/or transported with a cap on its stem to keep the stem free from dirt and debris. Prior to assembly, both caps are removed and an oil seal ring protector is positioned in the transmission opening configured to receive the drive shaft. The drive shaft is then inserted into the transmission opening and the oil seal ring protector is removed. Connection of the drive shaft to the transmission is then completed.

(3) Because multiple protective devices are involved, there are multiple opportunities in the current process for a cap or protector to be forgotten or inadvertently removed prior to assembly. Even with

the cap on the drive shaft stem, dirt and debris can still be introduced to the transmission via the drive shaft as the cap is not impervious to dirt and debris and dirt and debris can contaminate the shaft even if the cap is only removed for a short period of time.

SUMMARY

(4) Disclosed herein are implementations of a multipurpose protective cap for a transmission prior to and during assembly with a drive shaft. The multipurpose protective cap is multi-functional, protecting the transmission prior to assembly with the drive shaft, protecting the oil seal ring during insertion of the drive shaft, and removing debris from the drive shaft during insertion into the transmission

(5) A multipurpose protective cap for a transmission prior to and during assembly with a drive shaft as disclosed herein has a base portion having a cylindrical cross-section of a first inner diameter. The base portion comprises a side wall defining the cylindrical cross-section and having an outer edge and an inner edge, and a membrane spanning the cylindrical cross-section of the base portion at the outer edge of the side wall. An insertion portion extends from the base portion opposite the outer edge, the insertion portion having a cylindrical cross-section of a second inner diameter, the second inner diameter smaller than the first inner diameter. A contamination removal portion is formed in an interior surface of the insertion portion and configured to remove debris as the drive shaft passes through the insertion portion. A handle extends from the side wall of the base portion.

(6) Another implementation of a multipurpose protective cap is disclosed. The multipurpose protective cap is for a transmission prior to and during assembly with a drive shaft, the transmission having an opening for the drive shaft having an oil seal ring positioned around its inner circumference. The multipurpose protective cap comprises a base portion having a cylindrical cross-section of a first inner diameter, the first inner diameter being greater than an outer diameter of the oil seal ring. The base portion comprises a side wall defining the cylindrical cross-section and having an outer edge and an inner edge, and a membrane spanning the cylindrical cross-section of the base portion at the outer edge of the side wall. An insertion portion extends from the base portion proximate the inner edge, the insertion portion having a cylindrical cross-section of a second inner diameter, the second inner diameter smaller than the first inner diameter of the base portion. A contamination removal portion is formed in an interior surface of the insertion portion and configured to remove debris as the drive shaft passes through the insertion portion.

(7) Other implementations are contemplated based on the description herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIG. 1 is a schematic of a transmission coupled to a drive shaft.

(3) FIG. 2 is a schematic of at least a portion of a drive shaft.

(4) FIGS. 3A and 3B are perspective views of a multipurpose protective cap as disclosed herein.

(5) FIG. 4A is a cross-sectional view of the multipurpose protective cap disclosed herein inserted into an opening of the transmission.

(6) FIG. 4B is a cross-sectional view of the multipurpose protective cap inserted into the opening of the transmission as in FIG. 4A but with the drive shaft partially inserted.

(7) FIG. 5 is a schematic showing the multipurpose protective cap inserted into the opening of the transmission.

(8) FIG. 6 is a schematic showing the multipurpose protective cap inserted into the opening of the

- transmission with the drive shaft inserted through a membrane of the multipurpose protective cap.
- (9) FIG. 7 is a schematic showing the multipurpose protective cap removed from the opening of the transmission after the drive shaft has been inserted.
- (10) FIG. 8 is a plan view of the membrane of the multipurpose protective cap having weakened segments.
- (11) FIG. 9 is a plan view of the membrane of the multipurpose protective cap having alternating overlapping sections.
- (12) FIG. 10 is a cross-sectional view of the multipurpose protective cap showing the contamination removal portion.
- (13) FIG. 11 is a schematic illustrating the split and the corresponding concave and convex V-shaped profiles of the split in the multipurpose protective cap.

DETAILED DESCRIPTION

(14) FIG. 1 is a schematic of a transmission **10** connected to a drive shaft **20**. As used herein, “transmission” can be any transmission that is connected to a drive shaft, including but not limited to a continuously variable transmission, a power transfer unit and a final drive unit, and can be used in any type of vehicle. Conventionally, transmissions are stored and/or transported prior to assembly with a cap that seals the opening **12** in the transmission **10** configured to receive the drive shaft **20**. The drive shaft **20** is stored and/or transported with a cap on its stem **22** (shown in FIG. 2) to keep the stem **22** free from dirt and debris. Prior to assembly, both caps are removed and an oil seal ring protector is positioned in the transmission opening **12** configured to receive the drive shaft **20**. The oil seal ring protector protects the oil seal ring from damage as the drive shaft **20** is inserted into the opening **12** of the transmission **10**. The drive shaft **20** is inserted into the transmission opening **12** and the oil seal ring protector is removed. Connection of the drive shaft **20** to the transmission **10** is then completed, as illustrated in FIG. 1.

(15) Because multiple protective devices are involved, there are many opportunities in the conventional process for one or more protective devices to be forgotten, inadvertently removed prior to assembly or improperly installed. Even with the protective cap on the stem **22** of the drive shaft **20** during storage and/or transportation, dirt and debris can still be introduced to the transmission **10** via the stem **22** of the drive shaft **20** as the protective cap is not impervious to dirt and debris and dirt and debris can contaminate the stem **22** even if the cap is only removed for a short period of time. There are also multiple locations in which the protective devices are used. For example, in a front-wheel drive vehicle, there are two entry points in the transmission in which the drive shafts (front left and front right) are inserted. In an all-wheel drive vehicle, there are four locations, one each for the front left and front right drive shafts and one each for the rear left and rear right drive shafts.

(16) The multipurpose protective cap disclosed herein protects the transmission **10** during storage and/or transport and eliminates the need for the protective cap on the stem **22** of the drive shaft **20** as well as the oil seal ring protector. The multipurpose protective cap is designed to be inserted into the opening **12** of the transmission prior to storage and/or transportation and remain in the opening **12** until the drive shaft **20** is inserted into the opening **12**. The multipurpose protective cap can be used in each opening within a transmission, with two being used in a front-wheel drive vehicle and four being used in an all-wheel drive vehicle.

(17) An implementation of a multipurpose protective cap **100** for a transmission **10** prior to and during assembly with a drive shaft **20** is shown in FIGS. 3A and 3B. The multipurpose protective cap **100** has a base portion **102** having a cylindrical cross-section **104** of a first inner diameter D.sub.1. The cylindrical cross-section **104** is circular in this implementation. The base portion **102** comprises a side wall **106** defining the cylindrical cross-section **104** and having an outer edge **108** and an inner edge **110**, and a membrane **112** spanning the cylindrical cross-section **104** of the base portion **102** at the outer edge **108** of the side wall **106**. An insertion portion **120** extends from the base portion **102** opposite the outer edge **108**, the insertion portion **120** having a cylindrical cross-

section **122** of a second inner diameter $D_{sub.2}$, the second inner diameter $D_{sub.2}$ smaller than the first inner diameter $D_{sub.1}$. The insertion portion **120** essentially extends from the inner edge **110** of the side wall **106** of the base portion **102**, with a step **124** inward from the inner edge **110** to produce the smaller second inner diameter $D_{sub.2}$. A contamination removal portion **130** is formed in an interior surface **126** of the insertion portion **120** and configured to remove debris as the drive shaft **20** passes through the insertion portion **120**. A handle **140** extends from the side wall **106** of the base portion **102**.

(18) As illustrated in FIGS. **4A** and **4B**, the opening **12** in the transmission **10** for the drive shaft **20** has an inner circumference C around which an oil seal ring **30** is positioned. The oil seal ring **30** is configured to sealingly engage the stem **22** of the drive shaft **20** to prevent oil leakage. The first inner diameter $D_{sub.1}$ of the base portion **102** is greater than an outer diameter $D_{sub.3}$ of the oil seal ring **30**. In turn, the cylindrical cross-section **104** of the base portion **102** has a circumference that is greater than the circumference C of the opening **12** in the transmission **10**. These dimensions of the base portion **102** protect the entire oil seal ring **30** from damage during insertion of the drive shaft **20** and also prevent any dirt and debris from entering the opening **12** of the transmission **10**.

(19) As illustrated in FIGS. **4A** and **4B**, the second inner diameter $D_{sub.2}$ of the insertion portion **120** is sized to just receive the stem **22** of the drive shaft **20**. When the insertion portion **120** of the multipurpose protective cap **100** is inserted into the opening **12** of the transmission **10**, an outer surface **128** of the insertion portion **120** will contact the oil seal ring **30**, the contact with the oil seal ring **30** biasing the oil seal ring **30** to accommodate the insertion portion **120**. When the stem **22** is inserted and the multipurpose protective cap **100** removed, the oil seal ring **30** is unbiased enough to form a tight seal against the stem **22** of the drive shaft **20**.

(20) FIG. **5** shows the multipurpose protective cap **100** in place in the opening **12** of the transmission **10**. In FIG. **5**, an outer seal **40** can be seen, the outer seal configured to contact a wider portion of the drive shaft **20** when the drive shaft **20** is fully inserted. The multipurpose protective cap **100** can have a base portion **102** sized greater than the opening **12** but less than an outer circumference of the outer seal **40** as shown in FIG. **5** or can be sized greater than the opening **12** and greater than the outer circumference of the outer seal **40** as illustrated in FIGS. **4A** and **4B**.

(21) FIG. **4B** illustrates how the multipurpose protective cap **100** protects the oil seal ring **30** during insertion of the stem **22** of the drive shaft **20**. FIG. **6** illustrates the drive shaft **20** inserted so that the stem **22** is past the oil seal ring **30**. FIG. **7** illustrates the multipurpose protective cap **100** removed from the opening **12** of the transmission **10** and the drive shaft **20** fully inserted into the transmission **10**.

(22) The membrane **112** across the cylindrical cross-section **104** of the base portion **102** can be a thin, flexible material configured to be punctured with a predetermined force applied to the membrane **112** when the stem **22** of the drive shaft **20** is pushed against it during insertion of the drive shaft **20**. The membrane **112** can have weakened segments **114**, illustrated in FIG. **8** radiating from a center point **116** of the membrane **112** to facilitate the rupture of the membrane **112** upon the application of the predetermined force. Alternatively, the membrane **112** can comprise overlapping sections of flexible material. As shown in FIG. **9**, the overlapping sections can be the same size but, with the view in FIG. **9**, alternating sections **118** appear larger from the viewing side because they overlap alternating sections **118'**, which appear smaller from the viewing side. The overlap serves to ensure no dirt and debris enter the transmission during storage and/or transportation, keeping the opening **12** of the transmission **10** sealed.

(23) The width $W_{sub.1}$ of the insertion portion **120** is greater than a width $W_{sub.2}$ of the side wall **106** of the base portion **102** (shown in FIG. **4B**). The width $W_{sub.1}$ of the insertion portion **120** should be sufficient to completely cover the oil seal ring **30** and to provide sufficient surface area for the contamination removal portion **130**. The width $W_{sub.2}$ of the side wall **106** need only be wide enough to provide sufficient void space to allow for the membrane **112** to flex inwardly to either breaking point of the membrane or separation point of the membrane to allow for insertion of

the drive shaft **20**. This is schematically shown in FIG. **4B**.

(24) To ensure that the stem **22** of the drive shaft **20** does not introduce dirt and debris into the transmission **10**, a contamination removal portion **130** is formed on the interior surface **126** of the insertion portion **120**. The inclusion of the contamination removal portion **130** in the multipurpose protective cap **100** enables the elimination of the protective cap conventionally used on the stem **22** of the drive shaft **20** during storage/transportation.

(25) FIG. **10** shows a cross-section of the multipurpose protective cap **100** to better illustrate the contamination removal portion **130**. The contamination removal portion **130** can include one or both of a brush section **150** and a scraping section **160**. The brush section **150** is configured to remove dirt and debris from within the longitudinal ridges on the end of the stem **22** of the drive shaft, the ridges **24** illustrated in FIG. **4B**. The brush section **150** is positioned proximate or adjacent the inner edge **110** of the side wall **106** of the base portion **102**. The brush section **150** has elongated ridges **152** extending parallel with an insertion direction A, the elongated ridges **152** formed in the interior surface **126** around an inner circumference X of the insertion portion **120**. The elongated ridges **152** of the brush section **150** are sized to penetrate between ridges **24** of the stem **22**, so that as the stem **22** is inserted into the opening **12**, the elongated ridges **152** of the brush section **150** push out any dirt and debris between the ridges **24** of the stem **22**.

(26) The contamination removal portion **130** can include the scraping section **160** shown in FIG. **10**. The scraping section **160**, if used in conjunction with the brush section **150**, is positioned adjacent the brush section **150** opposite the inner edge **110** of the side wall **106** of the base portion **102**. The scraping section **160** is configured to scrape and collect dirt and debris during insertion from the stem **22** of the drive shaft **20** beyond the ridges **24**. If used with the brush section **150**, the scraping section **160** will collect the dirt and debris removed by the brush section **150**, preventing contamination of the transmission **10**. The scraping section **160** has an angled notch **162** formed in the interior surface **126** around the inner circumference X of the insertion portion **120**. The angled notch **162** is configured so that the apex **164** of the angled notch **162** contacts the stem **22**, scraping the stem **22** as it moves along the stem **22**. Dirt and debris collects in the valley **166** of the angled notch **162**. The scraping section **160** has one or more angled notches **162**. FIG. **10** illustrates six angled notches **162** formed at intervals I along the width W.sub.1 of the insertion portion **120** as a non-limiting example. The contamination removal portion **130** can extend the entire width W.sub.1 of the insertion portion **120** or may only extend along a portion of the width W.sub.1.

(27) As illustrated in FIG. **7**, when the drive shaft **20** is nearly completely inserted into the transmission **10**, the multipurpose protective cap **100** is removed, and the drive shaft **20** is fully inserted into the transmission **10**. To allow for removal of the multipurpose protective cap **100** after insertion of the drive shaft **20**, a split **170** is formed along the base portion **102** and insertion portion **120** along the insertion direction A, as shown in FIGS. **8** and **9**. The split **170** is shown formed along the base portion **102** and insertion portion **120** directly across from the handle **140**. The split **170** is not limited to this location and can be in another location. The split **170** can be aligned with a weakened segment **114** of the membrane **112** or at an overlap between alternating sections **118**, **118'** of the membrane **112**. One side **172** of the split **170** has a concave V-shaped profile **174** and another side **176** of the split **170** has a convex V-shaped profile **178** that is received in the concave V-shaped profile **174** when the split **170** is in a closed configuration. The snap fit of the concave and convex V-shaped profiles assist in keeping the split **170** in the closed configuration during storage/transportation and insertion of the drive shaft. Some pressure applied to the split **170** by pulling on the handle **140** separates the split **170** so that the multipurpose protective cap **100** can be removed from the opening **12** and from around the drive shaft **20** as illustrated in FIG. **7**.

(28) The handle **140** is configured to assist in removal of the multipurpose protective cap **100** after insertion of the drive shaft **20**. The handle **140** can be of any shape and size that fits within the space allotted by the opening's position and configuration within the transmission. The handle **140** can be a short portion extending from the base portion **102** sufficient to grasp with a figure and

thumb or can be a larger handle configured to grab with the entire hand.

(29) The multipurpose protective cap **100** can be integrally molded from rubber, plastic or resin. The membrane **112** can be of the same material as the remainder of the multipurpose protective cap **100**, but thinner to allow for the insertion of the drive shaft. Alternatively, the membrane **112** can be of a different material and fused with heat or welding, for example, to the base portion **102** of the multipurpose protective cap **100**.

(30) Unless defined otherwise, all technical and mechanical terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this disclosure belongs. Structures described herein are to be understood also to refer to functional equivalents of such structures.

(31) It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include the plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “an element” is a reference to one or more elements, and includes equivalents thereof known to those skilled in the art.

(32) While the disclosure has been described in connection with certain embodiments, it is to be understood that the disclosure is not to be limited to the disclosed embodiments but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. A multipurpose protective cap for a transmission prior to and during assembly with a drive shaft, the multipurpose protective cap comprising: a base portion having a cylindrical cross-section of a first inner diameter, the base portion comprising: a side wall defining the cylindrical cross-section and having an outer edge and an inner edge; and a membrane spanning the cylindrical cross-section of the base portion at the outer edge of the side wall; an insertion portion extending from the base portion opposite the outer edge, the insertion portion having a cylindrical cross-section of a second inner diameter, the second inner diameter smaller than the first inner diameter; a contamination removal portion formed in an interior surface of the insertion portion and configured to remove debris as the drive shaft passes through the insertion portion, wherein the contamination removal portion comprises a brush section formed in the interior surface; and a handle extending from the side wall of the base portion.
2. The multipurpose protective cap of claim 1, wherein the membrane is a flexible material configured to be punctured with a predetermined force.
3. The multipurpose protective cap of claim 2, wherein the membrane comprises weakened segments radiating from a center point of the membrane.
4. The multipurpose protective cap of claim 1, wherein the membrane comprises overlapping sections of flexible material.
5. The multipurpose protective cap of claim 1, wherein the brush section has elongated ridges extending parallel with an insertion direction, the elongated ridges formed in the interior surface around an inner circumference of the insertion portion.
6. The multipurpose protective cap of claim 1, wherein the brush section is positioned adjacent the inner edge of the side wall of the base portion, the contamination removal portion further comprising a scraping section, the scraping section having an angled notch formed in the interior surface around an inner circumference of the insertion portion, the scraping section positioned adjacent the brush section opposite the inner edge of the side wall of the base portion.
7. The multipurpose protective cap of claim 6, wherein the scraping section comprises a plurality of angled notches formed in the interior surface, each of the plurality of the angled notches formed at intervals along an insertion direction.

8. The multipurpose protective cap of claim 1, wherein a width of the insertion portion is greater than a width of the side wall of the base portion.
 9. The multipurpose protective cap of claim 1, further comprising: a split along an insertion direction of the multipurpose protective cap, wherein one side of the split has a concave V-shaped profile and another side of the split has a convex V-shaped profile that is received in the concave V-shaped profile when the split is in a closed configuration.
 10. A multipurpose protective cap for a transmission prior to and during assembly with a drive shaft, the multipurpose protective cap comprising: a base portion having a cylindrical cross-section of a first inner diameter, the base portion comprising: a side wall defining the cylindrical cross-section and having an outer edge and an inner edge; and a membrane spanning the cylindrical cross-section of the base portion at the outer edge of the side wall; an insertion portion extending from the base portion proximate the inner edge, the insertion portion having a cylindrical cross-section of a second inner diameter, the second inner diameter smaller than the first inner diameter of the base portion; and a contamination removal portion formed in an interior surface of the insertion portion and configured to remove debris as the drive shaft passes through the insertion portion, the contamination removal portion comprising a scraping section having an angled notch formed in the interior surface around an inner circumference of the insertion portion.
 11. The multipurpose protective cap of claim 10, wherein the membrane is a flexible material having weakened segments radiating from a center point of the membrane.
 12. The multipurpose protective cap of claim 10, wherein the membrane comprises overlapping sections of flexible material.
 13. The multipurpose protective cap of claim 10, wherein the contamination removal portion further comprises a brush section, the brush section having elongated ridges extending parallel with an insertion direction, the elongated ridges formed in the interior surface around an inner circumference of the insertion portion.
 14. The multipurpose protective cap of claim 13, wherein the brush section is positioned adjacent the inner edge of the side wall of the base portion, the scraping section positioned adjacent the brush section opposite the inner edge of the side wall of the base portion.
 15. The multipurpose protective cap of claim 10, wherein the scraping section comprises a plurality of angled notches formed in the interior surface, each of the plurality of the angled notches formed at intervals along an insertion direction.
 16. The multipurpose protective cap of claim 10, wherein a width of the insertion portion is greater than a width of the side wall of the base portion.
 17. The multipurpose protective cap of claim 10, further comprising: a split along an insertion direction of the multipurpose protective cap, wherein one side of the split has a concave V-shaped profile and another side of the split has a convex V-shaped profile that is received in the concave V-shaped profile when the split is in a closed configuration.
 18. The multipurpose protective cap of claim 10 further comprising: a handle extending from the side wall of the base portion.
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